Project Design Phase
Solution Architecture

Date	10 Feb 2026
Team ID	LTVIP2026TMIDS91131
Project Name	AutoSage App Using Gemini-Flash
Maximum Marks	4 Marks

Solution Architecture:

1. Overview

The AutoSage Solution Architecture is designed as a cloud-based, multimodal AI-driven system that integrates image processing and conversational AI to provide structured automotive insights. The architecture bridges the gap between fragmented vehicle information and intelligent decision-making by combining a web interface, AI processing engine, session management, and cloud-based AI services.

The system follows a modular layered architecture consisting of:

- Presentation Layer (User Interface)
- Application Layer (Business Logic)
- AI Processing Layer (Gemini 2.5 Flash)
- Session & Temporary Storage Layer
- Cloud Infrastructure Layer

This architecture ensures scalability, reliability, maintainability, and efficient data flow between users and AI services.

2. Architecture Components

2.1. User Interface Layer (Presentation Layer)

Technology: Streamlit (Python Web Framework)

Responsibilities:

- Accept vehicle image uploads
- Accept text-based queries
- Display structured vehicle reports
- Display conversational responses
- Manage mode switching (Image / Assistant)
- Show chat history

2.2. Application Logic Layer

Technology: Python

Modules:

- Image Validation & Processing
- Prompt Engineering Module
- Text Query Builder
- Response Formatter
- Error Handling Module
- Session State Management

Responsibilities:

- Construct structured prompts
- Handle input validation
- Process image bytes
- Format AI responses
- Maintain conversation history

2.3. AI Processing Layer

Technology: Google Gemini 2.5 Flash API

Responsibilities:

- Multimodal processing (Image + Text)
- Vehicle identification
- Structured report generation
- Budget-based recommendations
- Vehicle comparisons
- Maintenance advice
- EV suggestions

This is the core intelligence engine of the system.

2.4. Session & Temporary Storage Layer

Data Stores:

D1 – Session State

- Stores chat history
- Stores application mode (Image / Assistant)

D2 – Temporary Image Storage

- Holds uploaded image bytes temporarily during processing

No persistent database is currently implemented.

2.5. Cloud Infrastructure Layer

Deployment Options:

- Streamlit Cloud
- Local Server
- Cloud VM (AWS / GCP / Azure)

API Communication:

- Secure API key management via:
    - Streamlit Secrets (Cloud)
    - .env file (Local)

3. End-to-End Data Flow

Image-Based Analysis Flow

1. User uploads vehicle image
2. Image validated and converted to byte format
3. Structured prompt + image sent to Gemini API
4. Gemini processes multimodal input
5. AI returns structured vehicle report
6. Application formats and displays output

Conversational Assistant Flow

1. User enters vehicle-related query
2. Query structured into formatted prompt
3. Prompt sent to Gemini API
4. AI generates contextual response
5. Response stored in session state
6. Chat displayed with history

4. Architectural Characteristics

4.1 Modular Design

Separation between UI, logic, AI, and storage.

4.2 Scalable

AI processing handled by cloud-based Gemini API.

4.3 Secure

API keys managed via environment variables or secure cloud secrets.

4.4 Maintainable

Prompt engineering separated from UI logic.

4.5 Extensible

Future enhancements possible:

- Add database integration
- Add VIN recognition
- Add ownership cost estimation
- Add multilingual support
- Add dealership integration

5. Architecture Style

The system follows:

- Layered Architecture
- API-Centric Architecture
- Cloud-Based AI Integration
- Stateless Application with Session Memory

6. Solution Architecture Summary

The AutoSage architecture effectively transforms fragmented automotive research into a centralized AI-driven advisory experience. By leveraging cloud-based multimodal AI, structured prompt engineering, and lightweight web deployment, the solution ensures high usability, scalability, and maintainability while delivering intelligent vehicle insights in real time.

Solution Architecture Diagram:

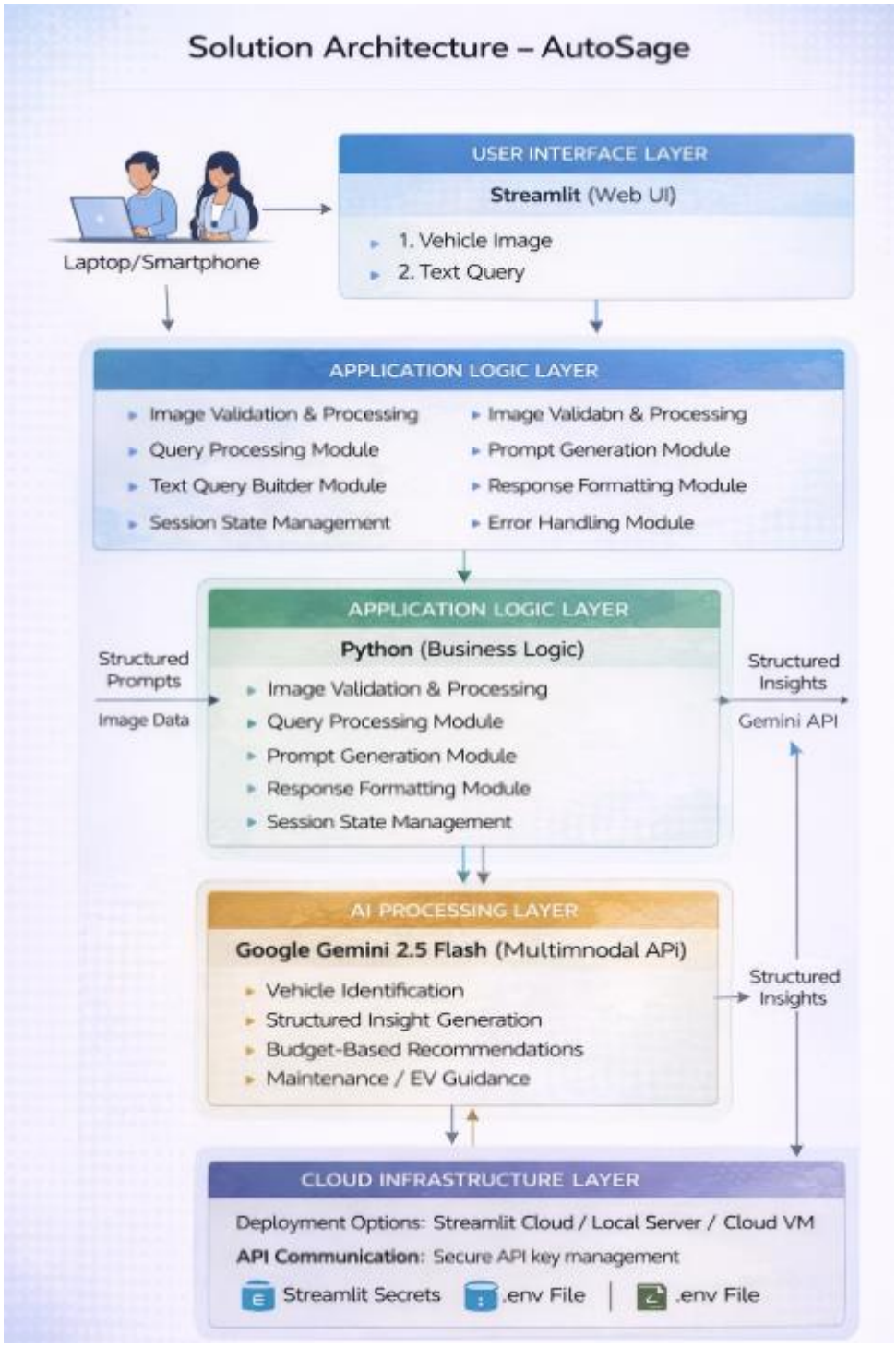

Figure 1: Architecture and data flow of the AutoSage Application